

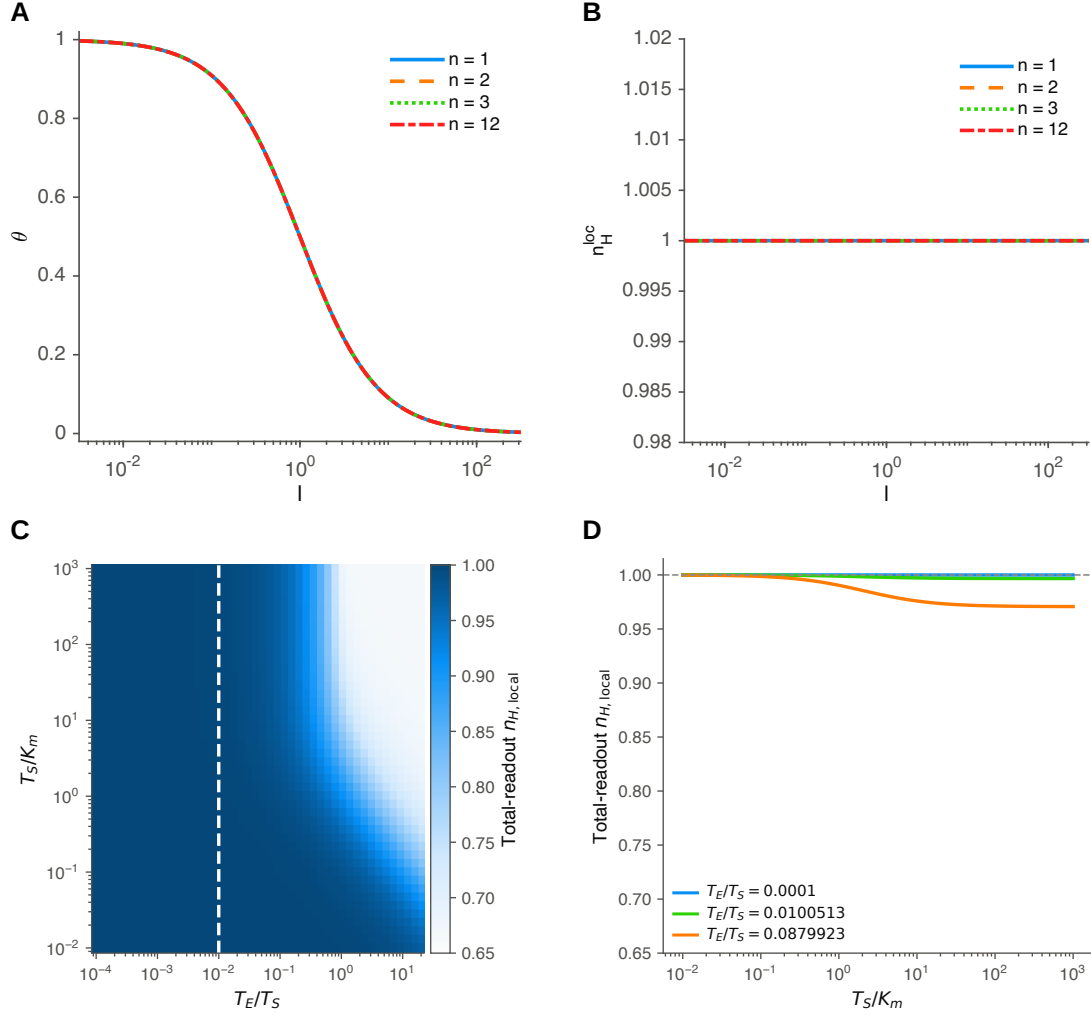


Figure S1: Independent-site invariance and the experimentally accessible readout. (A) Free-target mean modified fractions for $n=1, 2, 3$, and 12 identical independent sites coincide, with dimensionless input and unit midpoint. (B) Their local coefficients equal one throughout. (C) Local coefficient of the total-target readout at half-response across enzyme-to-target ratio and target saturation in the specified three-site model; the dashed line marks $T_E/T_S=0.01$. (D) Representative slices. For T_E/T_S at most 0.01 the coefficient remains at least 0.99745 across the plotted saturation range. Panels C and D are numerical calculations for the stated model; the exact invariant in A and B concerns free target.